Phase 1 Problem Statement: Counterfactual Conversation Emulation

Goal: estimate the causal deployment effect of replacing historical clinician final outputs with a fixed medical AI agent, using retrospective medical conversation data.

Problem Formulation

We study historical clinician-patient encounters where a clinician observed a patient context and wrote a final assessment and management plan. The target intervention is not a full multi-turn dialogue policy in v1. Instead, each encounter is one contextual-bandit decision: at time zero observe pre-decision context **X**, choose a free-form final clinical action **A** consisting of diagnosis/differential plus management plan, and evaluate a scalar rubric reward **Y in [0, 1]**.

The scientific question is: for encounters drawn from the same population as the logged clinician-note data, what would the expected quality score have been if the fixed agent policy had produced the final clinical output instead of the clinician?

Primary Estimand

```
V(pi_agent) = E_X [ E_{A ~ pi_agent(.|X)} [ q(X, A) ] ],   q(x, a) = E[Y(a) | X = x]
```

```
V(pi_b) = E_X [ E_{A ~ pi_b(.|X)} [ q(X, A) ] ],   Delta = V(pi_agent) − V(pi_b)
```

Interpretation: Delta is the average change in rubric-defined final clinical-output quality if the fixed agent replaced the clinician final output for the eligible historical encounter population. The logged behavior value is estimated by the empirical mean of clinician scores; the agent value requires off-policy evaluation because rewards are observed only for logged clinician actions.

Notation Table

Symbol	Meaning	Operational definition in this project
i	Encounter index	One historical medical conversation paired with a clinician note.
X_i	Patient context	Information available at time zero; excludes clinician final diagnosis, assessment, plan, and post-decision facts.
A_i^b	Behavior action	Clinician final diagnosis/differential and management plan extracted from the note.
pi_b	Behavior policy	Unknown human clinician policy that generated the logged action.
pi_agent	Target policy	Frozen model, prompt, input template, decoding settings, and tool policy.
Y_i	Observed reward	Continuous LLM-as-judge clinical quality score for (X_i, A_i^b), normalized to [0, 1].
phi(A)	Action representation	Embedding or cluster representation used for large-language-action OPE.
E_i	Embedded action	E_i = phi(A_i^b); target-agent actions also receive E_i^agent = phi(A_i^agent).

Identification Assumptions

- **Consistency:** observed clinician reward equals the potential reward under the observed action.
- **Time-zero validity:** X contains only pre-decision information.
- **Fixed target policy:** pi_agent is frozen before evaluation.
- **No interference:** one encounter's action does not affect another encounter's reward.
- **Conditional exchangeability:** given X, rubric reward for an action is comparable between logged clinician and counterfactual agent settings.

Estimator-Specific Assumptions

- **Common embedding support:** agent action embeddings lie in clinician-covered regions for comparable X.
- **MIPS no direct effect:** action effects on Y are mediated by phi(A); exact wording has no residual effect after conditioning on X and embedding.
- **OffCEM local correctness:** residual reward modeling captures within-cluster language effects sufficiently for doubly robust correction.
- **Rubric validity:** Y is a surrogate clinical-output quality measure, not a validated patient outcome.

Scope Decision

Exact sequence-level IPS is not viable because free-form language actions have near-zero exact overlap and the clinician behavior policy is unknown. The v1 benchmark therefore evaluates final-output replacement using embedding-space OPE and OffCEM-style robustness checks, while deferring full multi-turn dialogue-policy evaluation to future work.

Deliverable for onboarding Phase 1: one-page problem statement with estimand, assumptions, and notation table.